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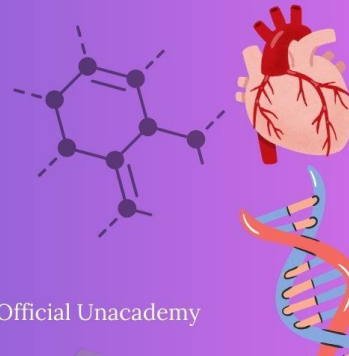
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Seep Pahuja Biology - Official Unacademy

LAST 10 YEARS

NEET PYQ



Chapter 9 : Biomolecules

1. The protein portion of an enzyme is called:

[NEET-2025]

- (1) Cofactor
- (2) Coenzyme
- (3) Apoenzyme
- (4) Prosthetic group

2. Match List I with List II:

[NEET-2025]

List-I

List-II

A. Adenosine

I. Nitrogen base

B. Adenylic acid

II. Nucleotide

C. Adenine

III. Nucleoside

D. Alanine

IV. Amino acid

Choose the option with all correct matches.

- (1) A-III, B-IV, C-II, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-III, C-I, D-IV

3. Which one of the following statements refers to Reductionist Biology?

[NEET-2025]

- (1) Physico-chemical approach to study and understand living organisms
- (2) Physiological approach to study and understand living organisms
- (3) Chemical approach to study and understand living organisms
- (4) Behavioural approach to study and understand living organisms

4. Which one of the following enzymes contains 'Haem' as the prosthetic group?

[NEET-2025]

- (1) RuBisCo
- (2) Carbonic anhydrase
- (3) Succinate dehydrogenase
- (4) Catalase

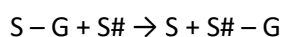
5. Histones are enriched with –

[NEET-2025]

- (1) Lysine & Arginine
- (2) Leucine & Lysine
- (3) Phenylalanine & Leucine
- (4) Phenylalanine & Arginine

6. Name the class of enzyme that usually catalyze the following reaction :

[NEET-2025]



Where, $G \rightarrow$ a group other than hydrogen

$S \rightarrow$ a substrate ; $S\# \rightarrow$ another substrate

- (1) Hydrolase
- (2) Lyase
- (3) Transferase
- (4) Ligase

7. Which of the following is a nucleotide?

[Re-NEET-2024]

- (1) Uridine
- (2) Adenylic acid
- (3) Guanine
- (4) Guanosine

8. Ligases is a class of enzymes responsible for catalysing the linking together of two compounds. Which of the following bonds is not catalysed by it?

[Re-NEET-2024]

- (1) C – C
- (2) P – O
- (3) C – O
- (4) C – N

9. Which of the following are not fatty acids?

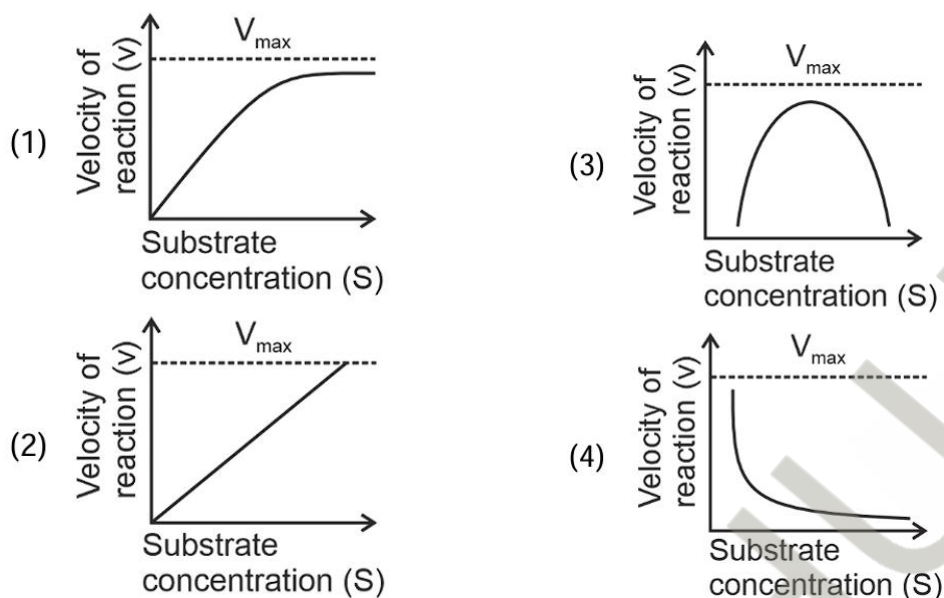
[Re-NEET-2024]

- | | |
|------------------|---------------------|
| A. Glutamic acid | B. Arachidonic acid |
| C. Palmitic acid | D. Lecithin |
| E. Aspartic acid | |

Choose the correct answer from the options given below :

- | | |
|---------------------|------------------|
| (1) C, D and E only | (2) A and B only |
| (3) A, D and E only | (4) B and C only |

10. Which of the following graphs depicts the effect of substrate concentration on velocity of enzyme catalysed reaction? [Re-NEET-2024]



11. Enzymes that catalyse the removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds, are known as : [Re-NEET-2024]

- (1) Transferases
- (2) Oxidoreductases
- (3) Dehydrogenases
- (4) Lyases

12. Match List-I with List-II.

[Re-NEET-2024]

List-I

List-II

A. Primary structure of protein

I. Human haemoglobin

B. Secondary structure of protein

II. Disulphide bonds

C. Tertiary structure of protein

III. Polypeptide chain

D. Quaternary structure of protein

IV. Alpha helix and β sheet

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-II, C-I, D-IV

(3) A-I, B-III, C-II, D-IV

(4) A-IV, B-III, C-II, D-I

13. Lecithin, a small molecular weight organic compound found in living tissues, is an example of: [NEET-2024]

- (1) Amino acids
- (2) Phospholipids
- (3) Glycerides
- (4) Carbohydrates

14. Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of: [NEET-2024]

- (1) Cofactor inhibition
- (2) Feedback inhibition
- (3) Competitive inhibition
- (4) Enzyme activation

15. The cofactor of the enzyme carboxypeptidase is: [NEET-2024]

- (1) Zinc
- (2) Niacin
- (3) Flavin
- (4) Haem

16. Match List-I with List-II [NEET-2024]

List-I	List-II
A. GLUT-4	I. Hormone
B. Insulin	II. Enzyme
C. Trypsin	III. Intercellular ground substance
D. Collagen	IV. Enables glucose transport into cells

Choose the correct answer from the options given below.

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-IV, C-I, D-II

17. Regarding catalytic cycle of an enzyme action, select the correct sequential steps : [NEET-2024]

- A. Substrate enzyme complex formation.
- B. Free enzyme ready to bind with another substrate.
- C. Release of products.
- D. Chemical bonds of the substrate broken.
- E. Substrate binding to active site.

Choose the correct answer from the options given below :

- (1) E, A, D, C, B
- (2) A, E, B, D, C
- (3) B, A, C, D, E
- (4) E, D, C, B, A

18. Inulin is a polymer of:

[NEET-2023-MANIPUR]

- (1) Amino acids
- (2) Glucose
- (3) Fructose
- (4) Galactose

19. Which of the following is not a secondary metabolite ?

[NEET-2023-MANIPUR]

- (1) Anthocyanin
- (2) Lecithin
- (3) Curcumin
- (4) Morphine

20. Match List-I with List-II.

[NEET -2023-MANIPUR]

List-I

List-II

- | | |
|----------------------------|--------------------------|
| (A) Protein | (I) C = C double bonds |
| (B) Unsaturated fatty acid | (II) Phosphodiester bond |
| (C) Nucleic acid | (III) Glycosidic bonds |
| (D) Polysaccharide | (IV) Peptide bonds |

Choose the correct answer from the options given below :

- (1) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (2) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)
- (3) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
- (4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

21. Match List-I with List-II.

[NEET -2023-MANIPUR]

List-I

List-II.

- | | |
|-----------------|---------------------|
| (A) Terpenoides | (I) Codeine |
| (B) Lectins | (II) Diterpenes |
| (C) Alkaloids | (III) Ricin |
| (D) Toxins | (IV) Concanavalin A |

Choose the correct answer from the options given below :

- (1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
- (2) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
- (3) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)
- (4) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

22. Cellulose does not form blue colour with Iodine because [NEET -2023]
- (1) It is a disaccharide
 - (2) It is a helical molecule
 - (3) It does not contain complex helices and hence cannot hold iodine molecules
 - (4) It breaks down when iodine reacts with it
23. Melonate inhibits the growth of pathogenic bacteria by inhibiting the activity of [NEET -2023]
- (1) Succinic dehydrogenase
 - (2) Amylase
 - (3) Lipase
 - (4) Dinitrogenase
24. The Exoskeleton of arthropods is composed of [NEET - 2022]
- (1) Cellulose
 - (2) Chitin
 - (3) Glucosamine
 - (4) Cutin
25. Read the following statements on lipids and find out correct set of statements: [NEET - 2022]
- (a) Lecithin found in the plasma membrane is a glycolipid
 - (b) Saturated fatty acids possess one or more $C = C$ bonds
 - (c) Gingelly oil has lower melting point, hence remains as oil in winter
 - (d) Lipids are generally insoluble in water but soluble in some organic solvents
 - (e) When fatty acid is esterified with glycerol, monoglycerides are formed
- Choose the correct answer from the options given below:
- (1) (a), (d) and (e) only
 - (2) (c), (d) and (e) only
 - (3) (a), (b) and (d) only
 - (4) (a), (b) and (c) only
26. A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose ? [NEET - 2022]
- (1) $C_{12}H_{24}O_{12}$
 - (2) $C_{12}H_{22}O_{11}$
 - (3) $C_{12}H_{24}O_{11}$
 - (4) $C_{12}H_{20}O_{10}$

27. Match List - I with List - II :

[RE-NEET - 2022]

List - I	List - II
(a) Adenine	(i) Pigment
(b) Anthocyanin	(ii) Polysaccharide
(c) Chitin	(iii) Alkaloid
(d) Codeine	(iv) Purine

Choose the correct answer from the options given below :

- (1) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)
- (2) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)
- (3) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)
- (4) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii)

28. Primary proteins are also called as polypeptides because :

[RE-NEET - 2022]

- (1) They are linear chains
- (2) They are polymers of peptide monomers
- (3) Successive amino acids are joined by peptide bonds
- (4) They can assume many conformations

29. In the enzyme which catalyses the breakdown of: $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$ the prosthetic group is : [RE-NEET - 2022]

- (1) Nicotinamide adenine dinucleotide
- (2) Haem
- (3) Zinc
- (4) Niacin

30. Given below are two statements:

[RE-NEET - 2022]

Statement I : Amino acids have a property of ionizable nature of $-\text{NH}_2$ and $-\text{COOH}$ groups, hence have different structures at different pH.

Statement II : Amino acids can exist as Zwitterionic form at acidic and basic pH.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement-I and Statement-II are correct
- (2) Both Statement-I and Statement-II are Incorrect
- (3) Statement-I is correct but Statement-II is incorrect
- (4) Statement-I is incorrect but Statement-II is correct

31. Which of the following are not secondary metabolites in plants ?

[NEET - 2021]

- (1) Morphine, codeine
- (2) Amino acids, glucose
- (3) Vinblastin, curcumin
- (4) Rubber, gums

32. Match List - I with List - II.

[NEET - 2021]

List - I

List - II

- | | |
|----------------------------|---------------------------|
| (a) Protein | (i) C = C double bonds |
| (b) Unsaturated fatty acid | (ii) Phosphodiester bonds |
| (c) Nucleic acid | (iii) Glycosidic bonds |
| (d) Polysaccharide | (iv) Peptide bonds |

(a) (b) (c) (d)

(1) (iv) (i) (ii) (iii)

(2) (i) (iv) (iii) (ii)

(3) (ii) (i) (iv) (iii)

(4) (iv) (iii) (i) (ii)

33. Identify the incorrect pair.

[NEET - 2021]

- (1) Alkaloids - Codeine
- (2) Toxin - Abrin
- (3) Lectins - Concanavalin A
- (4) Drugs - Ricin

34. Following are the statements with reference to 'lipids'.

[NEET - 2021]

- (a) Lipids having only single bonds are called unsaturated fatty acids.
- (b) Lecithin is a phospholipid.
- (c) Trihydroxy propane is glycerol.
- (d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
- (e) Arachidonic acid has 16 carbon atoms.

Choose the correct answer from the options given below.

- | | |
|----------------------|----------------------|
| (1) (a) and (b) only | (2) (c) and (d) only |
| (3) (b) and (c) only | (4) (b) and (e) only |

35. Identify the basic amino acid from the following.

[NEET-2020]

- (1) Valine
- (2) Tyrosine
- (3) Glutamic Acid
- (4) Lysine

36. Which of the following is the most abundant protein in the animals ?

[NEET-2020]

- (1) Insulin
- (2) Haemoglobin
- (3) Collagen
- (4) Lectin

37. Identify substances having glycosidic bond and peptide bond, respectively in their structure : [NEET-2020]

- (1) Inulin, insulin
- (2) Chitin, Cholesterol
- (3) Glycerol, trypsin
- (4) Cellulose, lecithin

38. Match the following :

- | | |
|----------------------------|--------------------|
| (a) Inhibitor of catalytic | (i) Ricin activity |
| (b) Possess peptide bonds | (ii) Malonate |
| (c) Cell wall material in | (iii) Chitin fungi |
| (d) Secondary metabolite | (iv) Collagen |

Choose the correct option from the following

- (a) (b) (c) (d)
- (1) (ii) (iii) (i) (iv)
- (2) (ii) (iv) (iii) (i)
- (3) (iii) (i) (iv) (ii)
- (4) (iii) (iv) (i) (ii)

39. Identify the statement which is incorrect.

[NEET - 2020 (COVID-19)]

- (1) Sulphur is an integral part of cysteine.
- (2) Glycine is an example of lipids.
- (3) Lecithin contains phosphorus atom in its structure.
- (4) Tyrosine possesses aromatic ring in its structure.

40. Which of the following glucose transporters is insulin-dependent ? [NEET - 2019]

- (1) GLUT I
- (2) GLUT II
- (3) GLUT III
- (4) GLUT IV

41. Concanavalin A is : [NEET - 2019]

- (1) an alkaloid
- (2) an essential oil
- (3) a lectin
- (4) a pigment

42. Consider the following statements : [NEET - 2019]

- (A) Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group.
- (B) A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.

Select the correct option.

- (1) Both (A) and (B) are true.
- (2) (A) is true and (B) is false.
- (3) Both (A) and (B) are false.
- (4) (A) is false and (B) is true.

43. Which of the following organic compounds is the main constituent of Lecithin ? [NEET - 2019(Odisha)]

- (1) Arachidonic acid
- (2) Phospholipid
- (3) Cholesterol
- (4) Phosphoprotein

44. "Ramachandran plot" is used to confirm the structure of :- [NEET - 2019(Odisha)]

- | | |
|-----------------------|--------------|
| (1) RNA | (2) Proteins |
| (3) Triacylglycerides | (4) DNA |

45. Prosthetic groups differ from co-enzymes in that :- [NEET - 2019(Odisha)]

- (1) they require metal ions for their activity.
- (2) they (prosthetic groups) are tightly bound to apoenzymes.
- (3) their association with apoenzymes is transient.
- (4) they can serve as co-factors in a number of enzyme-catalyzed reactions.

46. The two functional groups characteristic of sugars are [NEET - 2018]
- (1) hydroxyl and methyl
 - (2) carbonyl and methyl
 - (3) carbonyl and phosphate
 - (4) carbonyl and hydroxyl
47. Which of the following is an amino acid derived hormone ? [NEET - 2018]
- (1) Epinephrine
 - (2) Ecdysone
 - (3) Estradiol
 - (4) Estriol
48. Which of the following are not polymeric ? [NEET - 2017]
- (1) Proteins
 - (2) Polysaccharides
 - (3) Lipids
 - (4) Nucleic acids
49. Which one of the following statements is correct, with reference to enzymes ? [NEET - 2017]
- (1) Holoenzyme = Apoenzyme + Coenzyme
 - (2) Coenzyme = Apoenzyme + Holoenzyme
 - (3) Holoenzyme = Coenzyme + Co-factor
 - (4) Apoenzyme = Holoenzyme + Coenzyme
50. One of the major components of cell wall of most fungi is :- [NEET - I 2016]
- (1) Chitin
 - (2) Peptidoglycan
 - (3) Cellulose
 - (4) Hemicellulose
51. A typical fat molecule is made up of :- [NEET - I 2016]
- (1) Three glycerol molecules and one fatty acid molecule
 - (2) One glycerol and three fatty acid molecules
 - (3) One glycerol and one fatty acid molecule
 - (4) Three glycerol and three fatty acid molecules

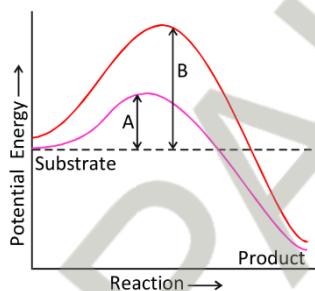
52. Which of the following is the least likely to be involved in stabilizing the three-dimensional folding of most proteins? [NEET - II 2016]

- (1) Hydrophobic interaction
- (2) Ester bonds
- (3) Hydrogen bonds
- (4) Electrostatic interaction

53. A non-proteinaceous enzyme is :- [NEET - II 2016]

- (1) Ligase
- (2) Deoxyribonuclease
- (3) Lysozyme
- (4) Ribozyme

54. Which of the following describes the given graph correctly? [NEET - II 2016]



- (1) Endothermic reaction with energy A in absence of enzyme and B in presence of enzyme
- (2) Exothermic reaction with energy A in absence of enzyme and B in presence of enzyme
- (3) Endothermic reaction with energy A in presence of enzyme and B in absence of enzyme
- (4) Exothermic reaction with energy A in presence of enzyme and B in absence of enzyme.

ANSWER KEY

1. (3)
2. (3)
3. (1)
4. (4)
5. (1)
6. (3)
7. (2)
8. (1)
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- 47. (3)
- 48. (4)
- 49. (1)
- 50. (3)
- 51. (4)
- 52. (4)
- 53. (4)
- 54. (4)

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